

Chapter 2

Basic Structures: Sets, Functions, Sequences, Sums, and Matrices

Section 2.1 :Sets
Section 2.2 :Set Operations

Sets

- ▶ A set is an unordered collection of objects, called elements or members of the set.
- ▶ A set is said to contain its elements. We write,
 - ▶ $a \in A$ to denote that a is an element of the set A .
 - ▶ The notation $a \notin A$ denotes that a is not an element of the set A .
- ▶ For example,
 - ▶ The notation $\{a, b, c, d\}$ represents the set with the four elements a, b, c , and d .



Sets(Contd.)

- ▶ Consider a set of all odd positive integers less than 10.

- ▶ **Roster Method:**

- ▶ Using this method, the set can be written as $\{1,3,5,7,9\}$

- ▶ **Set Builder Method**

- ▶ $O = \{x \mid x \text{ is an odd positive integer less than } 10\},$

Or,

- ▶ $O = \{x \in \mathbb{Z}^+ \mid x \text{ is odd and } x < 10\}.$



Sets(Contd.)

► Few common notations

N	$\{0, 1, 2, 3, \dots\}$, the set of natural numbers
Z	$\{\dots, -2, -1, 0, 1, 2, \dots\}$, the set of integers
Z^+	$\{1, 2, 3, \dots\}$, the set of positive integers
Q	$\{p/q \mid p \in Z, q \in Z, \text{ and } q \neq 0\}$, the set of rational numbers
R	the set of real numbers
R^+	the set of positive real numbers
C	the set of complex numbers
U	Universal Set



Equal Sets

► Equal Sets

- Two sets are equal if and only if they have the same elements.
- Therefore, if A and B are sets, then A and B are equal if and only if $\forall x(x \in A \leftrightarrow x \in B)$.
- We write $A = B$ if A and B are equal sets.
- The order of elements and the repetition of the same element in two sets don't matter in order for them to be equal.
- To show that two sets A and B are equal, show that $A \subseteq B$ and $B \subseteq A$.



Equal Sets(Contd.)

▶ **Example 1:**

- ▶ The sets $\{1, 3, 5\}$ and $\{3, 5, 1\}$ are equal, because they have the same elements.
- ▶ Similarly, the sets $\{1, 3, 5\}$ and $\{3, 5, 5, 5, 5, 1\}$ are equal, because they have the same elements.



Empty & Singleton Sets

▶ Empty Set

- ▶ A special set that has no elements. This set is called the *empty set*, or *null set*.
- ▶ It is denoted by $\emptyset$ or $\{\}$.

▶ Singleton Set

- ▶ A set with a single element is known as a singleton set.



Empty & Singleton Sets(Contd.)

But WAIT!!!



Empty & Singleton Sets(Contd.)

- ▶ Let, $A = \{ \}$ and $B = \{\emptyset\}$. Are both the sets empty?
- ▶ The Answer is **NO**.
 - ▶ Because set B has an element which is the empty element itself. Thus, B is a singleton set and not an empty set.



Venn Diagram

- ▶ **Venn Diagram:**
 - ▶ Graphical representation of sets.
 - ▶ In Venn diagrams the universal set U , which contains all the objects under consideration, is represented by a rectangle.
 - ▶ Inside this rectangle, circles or other geometrical figures are used to represent sets.
 - ▶ Sometimes points are used to represent the particular elements of the set.



Subsets

► Subsets

- The set A is a subset of B if and only if every element of A is also an element of B .
- The notation $A \subseteq B$ is used to indicate that A is a subset of the set B .
- Showing that A is a Subset of B
 - To show that $A \subseteq B$, show that if x belongs to A then x also belongs to B .
- Showing that A is Not a Subset of B
 - To show that $A \not\subseteq B$, find a single $x \in A$ such that $x \notin B$.
- Every nonempty set S is guaranteed to have at least two subsets, the empty set and the set S itself, that is, $\emptyset \subseteq S$ and $S \subseteq S$.



Cardinality

- ▶ **Size of a set**

- ▶ Let S be a set. If there are exactly n distinct elements in S where n is a nonnegative integer, we say that S is a finite set and that n is the cardinality of S . The cardinality of S is denoted by $|S|$.

- ▶ **Example:**

- ▶ Let A be the set of odd positive integers less than 10. Then $|A| = 5$.



Power Sets

► Power Sets

- Given a set S , the power set of S is the set of all subsets of the set S . The power set of S is denoted by $P(S)$.
- The cardinality of a power set of any set A is 2^n , where, n is the cardinality of A .
- Note that the empty set and the set itself are members of the set of subsets.



Power Sets(Contd.)

▶ **Example 1:**

- ▶ What is the power set of the set $\{0, 1, 2\}$?

▶ **Solution:**

- ▶ The power set $P(\{0, 1, 2\})$ is the set of all subsets of $\{0, 1, 2\}$.
Hence, $P(\{0, 1, 2\}) =$
 $\{\emptyset, \{0\}, \{1\}, \{2\}, \{0, 1\}, \{0, 2\}, \{1, 2\}, \{0, 1, 2\}\}.$

***3rd bracket for subsets



Power Sets(Contd.)

▶ **Example 2:**

- ▶ What is the power set of the empty set? What is the power set of the set $\{\emptyset\}$?

▶ **Solution:**

- ▶ The empty set has exactly one subset, namely, itself.
Consequently,

$$P(\emptyset) = \{\emptyset\}.$$

- ▶ The set $\{\emptyset\}$ has exactly two subsets, namely, $\emptyset$ and the set $\{\emptyset\}$ itself. Therefore,

$$P(\{\emptyset\}) = \{\emptyset, \{\emptyset\}\}.$$



Ordered Pairs

- ▶ The ordered n – *tuple* $(a_1, a_2, \dots, a_n)$ is the ordered collection that has a_1 as its first element, a_2 as its second element,..., and a_n as its n^{th} element.
- ▶ We say that two ordered n -tuples are equal if and only if each corresponding pair of their elements is equal.
 - ▶ In other words, $(a_1, a_2, \dots, a_n) = (b_1, b_2, \dots, b_n)$ if and only if $a_i = b_i$, for $i = 1, 2, \dots, n$.
- ▶ In particular, ordered 2-tuples are called ordered pairs.
 - ▶ The ordered pairs (a, b) and (c, d) are equal if and only if $a = c$ and $b = d$. Note that (a, b) and (b, a) are not equal unless $a = b$.



Ordered Pairs(Contd.)

▶ **Example I:**

- ▶ What are the ordered pairs in the set R having the less than or equal to relation, which contains (a, b) if $a \leq b$, on the set $\{0, 1, 2, 3\}$?

▶ **Solution:**

The ordered pair (a, b) belongs to R if and only if both a and b belong to $\{0, 1, 2, 3\}$ and $a \leq b$. Consequently, the ordered pairs in R are

$(0,0), (0,1), (0,2), (0,3), (1,1), (1,2), (1,3), (2,2), (2,3),$ and $(3,3)$.



Cartesian Products

- ▶ Let A and B be two sets.
- ▶ The Cartesian product of A and B , denoted by $A \times B$, is the set of all ordered pairs (a, b) , where $a \in A$ and $b \in B$. Hence,

$$A \times B = \{(a, b) \mid a \in A \wedge b \in B\}.$$



Cartesian Products(Contd.)

▶ **Example I:**

- ▶ Let A represent the set of all students at a university, and let B represent the set of all courses offered at the university. What is the Cartesian product $A \times B$ and how can it be used?

▶ **Solution:**

The Cartesian product $A \times B$ consists of all the ordered pairs of the form (a, b) , where a is a student at the university and b is a course offered at the university. One way to use the set $A \times B$ is to represent all possible enrollments of students in courses at the university.



Cartesian Products(Contd.)

▶ **Example 2:**

- ▶ What is the Cartesian product of $A = \{1, 2\}$ and $B = \{a, b, c\}$?

▶ **Solution:**

The Cartesian product $A \times B$ is

$$A \times B = \{(1, a), (1, b), (1, c), (2, a), (2, b), (2, c)\}.$$

***First bracket for cartesian products



Cartesian Products(Contd.)

► **Example 3:**

- Show that the Cartesian product $B \times A$ is not equal to the Cartesian product $A \times B$, where A and B are $A = \{1, 2\}$ and $B = \{a, b, c\}$.

► **Solution:**

The Cartesian product $A \times B$ is

$$A \times B = \{(1, a), (1, b), (1, c), (2, a), (2, b), (2, c)\}.$$

The Cartesian product $B \times A$ is

$$B \times A = \{(a, 1), (a, 2), (b, 1), (b, 2), (c, 1), (c, 2)\}.$$

Since Cartesian Product is the set of all ordered pairs, thus by definition of ordered pairs we can conclude that the Cartesian product $B \times A$ is not equal to the Cartesian product $A \times B$.



Section 2.2: Set Operations



Union of Sets

- ▶ Let A and B be sets.
- ▶ The union of the sets A and B , denoted by $A \cup B$, is the set that contains those elements that are either in A or in B , or in *both*.
- ▶ An element x belongs to the union of the sets A and B if and only if x belongs to A or x belongs to B . This tells us that

$$A \cup B = \{x \mid x \in A \vee x \in B\}.$$



Union of Sets(Contd.)

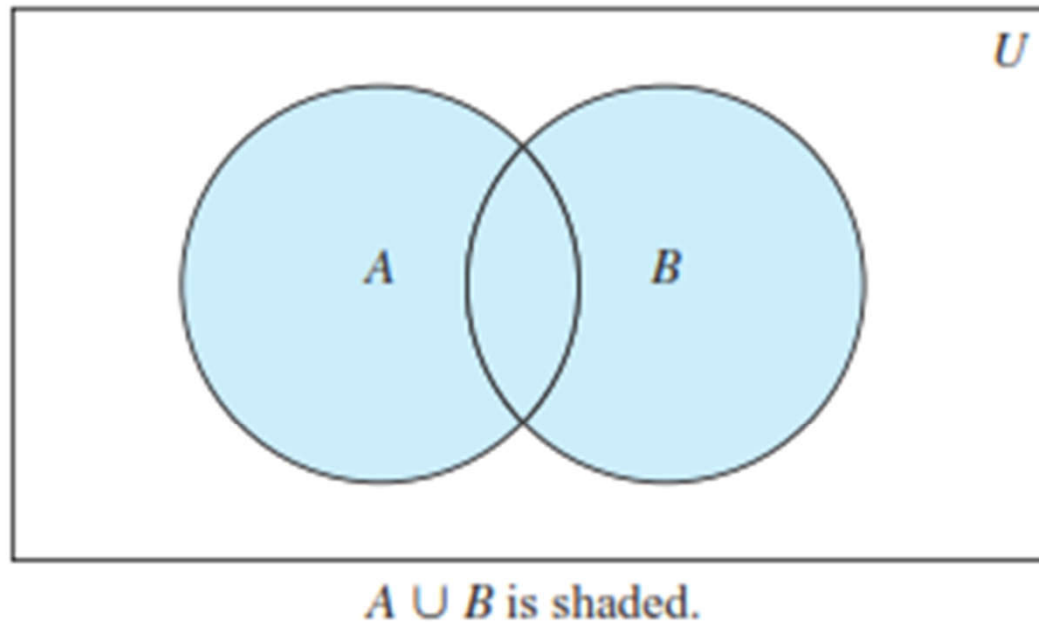


FIGURE 1 Venn Diagram of the Union of A and B .



Union of Sets(Contd.)

▶ **Example 1:**

- ▶ Find the union of the sets $\{1, 3, 5\}$ and $\{1, 2, 3\}$.

▶ **Solution:**

The union of the two sets $\{1, 3, 5\}$ and $\{1, 2, 3\}$ can be written as,

$$\{1, 3, 5\} \cup \{1, 2, 3\} = \{1, 2, 3, 5\}$$

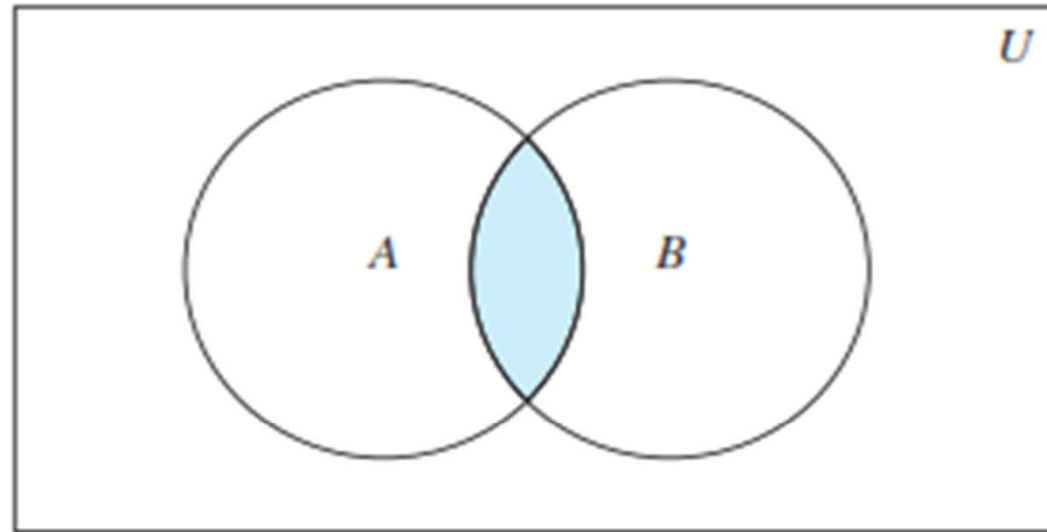


Intersection of Sets

- ▶ Let A and B be sets.
- ▶ The intersection of the sets A and B , denoted by $A \cap B$, is the set containing those elements in both A and B .
- ▶ An element x belongs to the intersection of the sets A and B if and only if x belongs to A and x belongs to B .
This tells us that



Intersection of Sets(Contd.)



$A \cap B$ is shaded.

FIGURE 2 Venn Diagram of the Intersection of A and B .



Intersection of Sets(Contd.)

► **Example I:**

- Find the intersection of the sets $\{1, 3, 5\}$ and $\{1, 2, 3\}$.

► **Solution:**

The intersection of the two sets $\{1, 3, 5\}$ and $\{1, 2, 3\}$ can be written as,

$$\{1, 3, 5\} \cap \{1, 2, 3\} = \{1, 3\}$$



Disjoint Sets

- ▶ Two sets are called disjoint if their intersection is the empty set.

- ▶ **Example I:**

- ▶ Let $A = \{1, 3, 5, 7, 9\}$ and $B = \{2, 4, 6, 8, 10\}$.
Because $A \cap B = \emptyset$, A and B are disjoint.



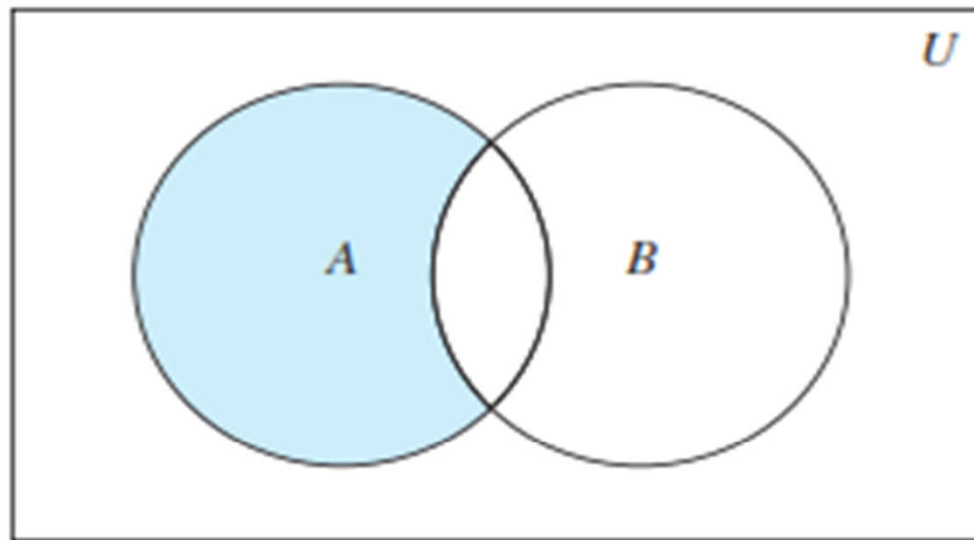
Difference of Sets

- ▶ Let A and B be sets.
- ▶ The difference of A and B , denoted by $A - B$, is the set containing those elements that are in A but not in B . The difference of A and B is also called the complement of B with respect to A .
- ▶ The difference of sets A and B is sometimes denoted by $A \setminus B$.
- ▶ An element x belongs to the difference of A and B if and only if $x \in A$ and $x \notin B$. This tells us that

$$A - B = \{x \mid x \in A \wedge x \notin B\}.$$



Difference of Sets(Contd.)



$A - B$ is shaded.

FIGURE 3 Venn Diagram for the Difference of A and B .



Difference of Sets(Contd.)

▶ **Example 1:**

- ▶ Find the difference of the sets $\{1, 3, 5\}$ and $\{1, 2, 3\}$.

▶ **Solution:**

The difference of the two sets $\{1, 3, 5\}$ and $\{1, 2, 3\}$ can be written as,

$$\{1, 3, 5\} - \{1, 2, 3\} = \{5\}$$

**elements that is in A but not in B



Complement of Sets

- ▶ Let U be the universal set. The complement of the set A , denoted by A , is the complement of A with respect to U . Therefore, the complement of the set A is $U - A$.
- ▶ An element belongs to A if and only if $x \notin A$. This tells us that

$$A = \{x \in U \mid x \notin A\}$$



Complement of Sets(Contd.)

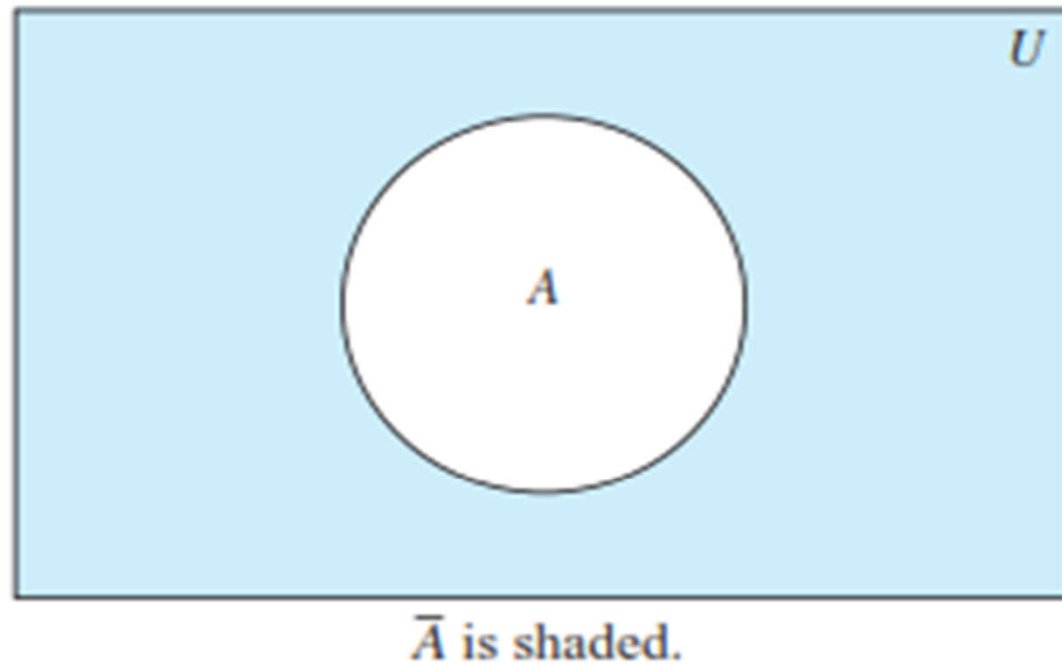


FIGURE 4 Venn Diagram for
the Complement of the Set A .



Complement of Sets(Contd.)

► **Example I:**

- Let $A = \{a, e, i, o, u\}$ (where the universal set is the set of letters of the English alphabet). Then

$$\bar{A} = \{b, c, d, f, g, h, j, k, l, m, n, p, q, r, s, t, v, w, x, y, z\}.$$



Set Identities

TABLE 1 Set Identities.

<i>Identity</i>	<i>Name</i>
$A \cap U = A$ $A \cup \emptyset = A$	Identity laws
$A \cup U = U$ $A \cap \emptyset = \emptyset$	Domination laws
$A \cup A = A$ $A \cap A = A$	Idempotent laws
$\overline{\overline{A}} = A$	Complementation law
$A \cup B = B \cup A$ $A \cap B = B \cap A$	Commutative laws
$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$ $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$	Associative laws
$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$ $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$	Distributive laws
$\overline{A \cap B} = \overline{A} \cup \overline{B}$ $\overline{A \cup B} = \overline{A} \cap \overline{B}$	De Morgan's laws
$A \cup (A \cap B) = A$ $A \cap (A \cup B) = A$	Absorption laws
$A \cup \overline{A} = U$ $A \cap \overline{A} = \emptyset$	Complement laws

Set Identities(Contd.)

► Example I:

► Prove that $\overline{A \cap B} = \bar{A} \cup \bar{B}$.

► Solution:

Let,

$$\begin{aligned}x \in \overline{A \cap B} &\equiv x \notin (A \cap B) \\&\equiv x \notin A \text{ or } x \notin B \\&\equiv x \in \bar{A} \text{ or } x \in \bar{B} \\&\equiv x \in (\bar{A} \cup \bar{B})\end{aligned}$$

$$\therefore \overline{A \cap B} \subseteq \bar{A} \cup \bar{B}$$

Again Let,

$$\begin{aligned}x \in (\bar{A} \cup \bar{B}) &\equiv x \in \bar{A} \text{ or } x \in \bar{B} \\&\equiv x \notin A \text{ or } x \notin B \\&\equiv x \notin (A \cap B) \\&\equiv x \in \overline{A \cap B}\end{aligned}$$

$$\therefore \bar{A} \cup \bar{B} \subseteq \overline{A \cap B}$$

$$\therefore \overline{A \cap B} = \bar{A} \cup \bar{B}$$

Set Identities(Contd.)

► **Example 2:**

- Use set builder notation and logical equivalences to establish the first De Morgan law $A \cap B = \bar{A} \cup \bar{B}$.

► **Solution:**

$$\begin{aligned}\overline{A \cap B} &= \{x \mid x \notin A \cap B\} \\ &= \{x \mid \neg(x \in (A \cap B))\} \\ &= \{x \mid \neg(x \in A \wedge x \in B)\} \\ &= \{x \mid \neg(x \in A) \vee \neg(x \in B)\} \\ &= \{x \mid x \notin A \vee x \notin B\} \\ &= \{x \mid x \in \bar{A} \vee x \in \bar{B}\} \\ &= \{x \mid x \in \bar{A} \cup \bar{B}\} \\ &= \bar{A} \cup \bar{B}\end{aligned}$$

Computer Representation of Sets

- ▶ Assume that a universal set U is finite (and of reasonable size so that the number of elements of U is not larger than the memory size of the computer being used).
- ▶ First, specify an arbitrary ordering of the elements of U , for instance $a_1, a_2, \dots, a_n$.
- ▶ Secondly, represent a subset A of U with the bit string of length n , where the i^{th} bit in this string is 1 if a_i belongs to A and is 0 if a_i does not belong to A .



Computer Representation of Sets(Contd.)

▶ **Example 1:**

- ▶ Let $U = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$, and the ordering of elements of U has the elements in increasing order; that is, $a_i = i$.
 - ▶ What bit strings represent the subset of all odd integers in U ?
 - ▶ What bit strings represent the subset of all even integers in U ?
 - ▶ What bit strings represent the subset of integers not exceeding 5 in U ?



Computer Representation of Sets(Contd.)

► **Solution:**

The bit string that represents the set of odd integers in U , namely, $\{1, 3, 5, 7, 9\}$, has a one bit in the *first, third, fifth, seventh and ninth* positions, and a *zero* elsewhere. It is

10 1010 1010.

Similarly, we represent the subset of all even integers in U , namely, $\{2, 4, 6, 8, 10\}$, by the string

01 0101 0101.

The set of all integers in U that do not exceed 5, namely, $\{1, 2, 3, 4, 5\}$, is represented by the string

11 1110 0000.



Computer Representation of Sets(Contd.)

▶ **Example 2:**

▶ Let $U = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$, and the ordering of elements of U has the elements in increasing order; that is, $a_i = i$.

- ▶ What bit strings represent the complement of the subset of all odd integers in U ?



Computer Representation of Sets(Contd.)

► **Solution:**

The bit string that represents the set of odd integers in U , namely, $\{1, 3, 5, 7, 9\}$, has a one bit in the *first, third, fifth, seventh and ninth* positions, and a *zero* elsewhere. It is

10 1010 1010.

The bit string for the complement of this set is obtained by replacing 0s with 1s and vice versa. This yields the string

01 0101 0101

which corresponds to the set $\{2, 4, 6, 8, 10\}$.



Computer Representation of Sets(Contd.)

▶ **Example 3:**

- ▶ The bit strings for the sets $\{1, 2, 3, 4, 5\}$ and $\{1, 3, 5, 7, 9\}$ are 11 1110 0000 and 10 1010 1010, respectively. Use bit strings to find the union and intersection of these sets.

▶ **Solution:**

The bit string for the union of these sets is,

$11\ 1110\ 0000 \vee 10\ 1010\ 1010 = 11\ 1110\ 1010$,
which corresponds to the set $\{1, 2, 3, 4, 5, 7, 9\}$.

The bit string for the intersection of these sets is,

$11\ 1110\ 0000 \wedge 10\ 1010\ 1010 = 10\ 1010\ 0000$,
which corresponds to the set $\{1, 3, 5\}$.



THE END

